PROJECT DESIGN PHASE-II

Data Flow Diagram & User Stories

Data Flow Diagrams

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

The Data Flow Diagram (DFD) illustrates the movement of data through the Flood Prediction System. The process begins when an authorized user, such as a meteorologist or disaster management officer, enters weather-related parameters including annual rainfall, seasonal rainfall, and cloud visibility through the Flask web interface. The system validates and preprocesses the input data by converting it into the format required by the machine learning model.

The processed data is then passed to the trained prediction model, which may use Decision Tree, Random Forest, K-Nearest Neighbours (KNN), or XGBoost algorithms. Based on the learned patterns from historical weather data, the model predicts whether there is a high or low risk of flooding. Finally, the prediction result is displayed through the web application. The output enables authorities to make timely decisions regarding early warning alerts, evacuation planning, disaster response, and resource allocation. This data flow ensures quick, accurate, and user-friendly flood risk assessment suitable for real-time disaster management.

MACHINE LEARNING PIPELINE & DATA FLOW ARCHITECTURE

1	2	3	4	5	6	7
Data Sources	**Ingestion & Schema**	**Data Cleaning**	**Transformation**	**Feature Eng.**	**Dim. Reduction**	**Train & Eval**
• Tabular DB / Warehouse • CSV / Parquet Files • APIs / Streams	• Type coercion & mapping • Deduplication • Basic QC checks	• Missing-value strategy • Outlier handling • Consistency checks	• Standardise/ Normalise • Power/ Quantile transforms • One-hot encoding	• Domain interactions • AutoFE libraries • Selection filters	• PCA / UMAP / t-SNE • Autoencoders / VAE	• CV / hold-out split • Model training • Metrics evaluation